AI Image Generator

10. Video and Music

Version 2.2
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Platform: NVIDIA GeForce RTX 3090, Windows 11, Python 3.12

Images, video and music on one GPU: Z-Image Turbo, SDXL fine-tunes, Qwen-Image, FLUX.1, LoRAs, LTX-Video, Wan 2.1, MusicGen and Real-ESRGAN, served locally with Gradio in the Frazile AI studio UI, with per-image GPU telemetry recorded in SQLite and an Admin panel that manages models, LoRAs and downloads and shows live system statistics.

10. Video and Music

10.1 Overview

Version 2.2 adds two model families next to the image families. They are ordinary registry entries, so the Admin panel lists, edits, downloads, loads and unloads them like any image model, and the Add tab accepts any LTX-Video, Wan, MusicGen or Stable Audio repository. They are used from their own tabs because their controls (length, frame rate, start image, temperature) have no meaning for images.

Model	Key	Family	Loader	Output	Limits	Download
LTX-Video 2B (Lightricks)	ltx	video	ltx	24 fps H.264 MP4	2 to 10 s; 768x512, 512x768, 832x480, 480x832, 1024x576, 576x1024; text- and image-to-video	28 GB
Wan 2.1 T2V 1.3B (Wan-AI)	wan	video	wan	16 fps H.264 MP4	2 to 6 s; same sizes, 832x480 native; text-to-video only	29 GB
MusicGen stereo medium (Meta)	musicgen	music	musicgen	32 kHz stereo WAV	3 to 30 s, instrumental	4.1 GB
MusicGen stereo small	musicgen-small	music	musicgen	32 kHz stereo WAV	3 to 30 s, instrumental	1.3 GB
Stable Audio Open 1.0 (Stability AI)	stable-audio	music	stable_audio	44.1 kHz stereo WAV	3 to 47 s; gated repo, needs a Hugging Face token; untested on the development machine	5.3 GB

The loader column is the architecture media.py detects from the repository (model_index.json for diffusers pipelines, config.json model_type for MusicGen). An admin-added repo gets the same treatment, so a fine-tuned LTX or a MusicGen melody checkpoint works without code changes.

10.2 The Video tab

- Prompt and chips: the model responds to a short description of scene, motion and camera; the chips add tracking shot, drone shot, close-up, time-lapse, slow motion, golden hour, cinematic lighting, rain and fog.
- Model box: LTX-Video 2B or Wan 2.1 (downloaded models first). Changing it resets steps, guidance, the negative prompt, the maximum length and shows or hides the start-image box.
- Size: six presets that both models accept (multiples of 32 for LTX, 16 for Wan). Length: 2 to 10 s; the frame count is rounded to 8k+1 (LTX) or 4k+1 (Wan) frames at the model's frame rate, so 4 s is 97 frames with LTX and 65 with Wan.
- Steps and guidance: LTX 40 and 3.0, Wan 30 and 5.0 by default (editable per model in Admin). Seed: -1 picks a random one; the clip's seed is in its details and History row.
- Start image (LTX only): any picture, including a Studio result dragged in. It is centre-cropped and resized to the chosen size and passed to the image-to-video pipeline, which shares the loaded components.
- Result: the clip plays in the result box; the details report lists timing, size, frames, fps, length, seed, file and the GPU telemetry.
- Upscale video: Full HD (1920), 2K (2560) or 4K (3840) long side. Frames are read back from the MP4, run one by one through Real-ESRGAN's compact video network (realesr-general-x4v3, 1.2M parameters, fetched into models/ on first use; the x4plus image network is the fallback) with a single x4 pass and a Lanczos resample to the exact target, and streamed into a new MP4 at crf 18. A sound track already added is kept.
- Add soundtrack: the box lists the tracks made in the Music tab; the chosen one is muxed under the current clip, cut to its length, with a one-second fade-out, into a new _music.mp4.
- Clear resets the tab; the clip stays in outputs/ and in History.

10.3 The Music tab

- Prompt and chips: genre, mood, instruments and tempo; chips for lo-fi hip hop, cinematic orchestral, synthwave, acoustic folk, jazz trio, techno, ambient pads, epic trailer, upbeat, melancholic, relaxing and 120 bpm.
- Length 3 to 30 s with MusicGen (47 s with Stable Audio); guidance (3 by default) pulls the result towards the text; temperature (1.0) adds variety; seed as in the Video tab.
- MusicGen generates 50 tokens per second of audio autoregressively, so progress is reported per token; a 15 s track is 750 decoder steps.
- The negative prompt box appears only for Stable Audio; MusicGen has no such input.
- Result: a player with the track and the details report; the file is a 16-bit WAV in outputs/.

10.4 Sharing one GPU

The image ModelManager and the MediaManager share one lock, so one job runs at a time, and media.prepare_for() decides what must leave the card before a model is used:

About to use	Image model loaded	Video model loaded	Music model loaded
An image model	kept (swapped as usual)	unloaded	kept, unless the image model is Z-Image, Qwen-Image or FLUX
A video model	unloaded	swapped if different	unloaded
A music model	kept while free VRAM >= need + 2 GB, else unloaded	unloaded	swapped if different

The next image generation reloads the image model automatically, so the user never unloads anything by hand. The header pill and the Loaded now line show both managers, and Admin > Models > Unload frees both.

10.5 How the models are loaded

- The repositories ship their text encoders in fp32: T5-XXL (LTX) is 19 GB and UMT5-XXL (Wan) 23 GB on disk, more than this machine's free RAM. Every part is therefore loaded with device_map onto the GPU shard by shard in bf16, never assembled on the CPU. The first load saves a bf16 copy of the text encoder under models/cache/, which halves later load times.
- LTX: T5EncoderModel + LTXVideoTransformer3DModel + AutoencoderKLLTXVideo with tiling, about 14 GB resident. Wan: UMT5EncoderModel in 8-bit on cards under 32 GB (its 11 GB bf16 form plus the fp32 VAE decode paged VRAM on the 24 GB card and took over ten minutes) + WanTransformer3DModel + fp32 AutoencoderKLWan with tiling and the UniPC scheduler (flow_shift 3.0).
- MusicGen: MusicgenForConditionalGeneration in fp16 with its processor, 4.1 GB resident; Stable Audio: StableAudioPipeline in fp16.
- Downloads fetch only the diffusers sub-folders (model_index.json and */*). This matters for LTX, whose repository also holds about 250 GB of single-file checkpoints for other runtimes.

10.6 Files, thumbnails and the database

- Clips: outputs/<timestamp>_seed<seed>.mp4; upscales add _up<px>; soundtracks add _music. Tracks: outputs/<timestamp>_seed<seed>.wav.
- Thumbnails in outputs/.thumbs/: the first frame (JPEG) for clips, a waveform picture (PNG) for tracks.
- generations.db rows carry kind = video or music plus duration_s, fps, frames, sample_rate and soundtrack_path; the Video tab's upscale fills the usual upscale columns. The History tab filters by kind and shows the right player; Reuse prompt returns to the Video or Music tab.

10.7 Measured on the RTX 3090

Operation	Settings	Time	VRAM
LTX-Video 2B, first load	fp32 to bf16 conversion of T5-XXL, bf16 copy written to models/cache/	186 s	14.2 GB resident
LTX-Video 2B, later loads	from the bf16 copy (inside the app)	94 s	14.2 GB resident
LTX text-to-video	768x512, 89 frames (3.7 s), 30 steps, through the app	29.9 s + 1 s encode	15.5 GB peak
LTX text-to-video	768x512, 49 frames (2 s), 20 steps	11.4 s	17.0 GB reserved
LTX image-to-video	768x512, 65 frames (2.7 s), 20 steps, from a 4K Studio PNG, through the app	15.7 s	14.9 GB peak
LTX image-to-video	768x512, 49 frames, 20 steps	12.9 s	15.3 GB peak
Wan 2.1 1.3B, first load	UMT5-XXL converted to bf16 and cached, then quantised to 8-bit	266 s	11.1 GB resident
Wan 2.1 1.3B, later loads	bf16 copy quantised to 8-bit	164 s	11.1 GB resident
Wan text-to-video	832x480, 49 frames (3 s), 30 steps, UniPC	205 s (about 6 s per step)	12.9 GB peak
Video upscale to 4K	65 frames 768x512 to 3840x2560, compact Real-ESRGAN, one pass + resample, H.264 crf 18	49 s (0.76 s per frame)	under 1 GB extra
Video upscale to 4K (fallback)	same clip with the x4plus image network	135 s one pass, 204 s two passes	under 1 GB extra
Soundtrack mux	10 s WAV under a 2.7 s clip, AAC, fade-out	0.4 s	none
MusicGen stereo medium, load	fp16, from the Hugging Face cache	24 to 27 s	4.1 GB resident
MusicGen stereo medium, track	10 s, guidance 3, temperature 1.0, through the app	36 s	4.2 GB peak
MusicGen stereo medium, track	6 s, guidance 3	23.6 s	4.6 GB reserved
Image after a video	Juggernaut XL v9 reloaded automatically after LTX-Video was unloaded	31 s load + 5.0 s image	10.4 GB peak

Measured on 22 October 2026 with torch 2.14.1, diffusers 0.41 and transformers 5.20. Video generation time grows roughly linearly with frames x steps; 768x512 at 97 frames and 40 steps is the recommended quality setting and takes about four times the 49-frame, 20-step figure.

10.8 Limits and tips

- Clips are short by design (2 to 10 s). For longer sequences, generate the next clip from the last frame of the previous one (save a frame, use it as the start image) and join the files with ffmpeg.
- LTX responds to specific, literal descriptions of motion; abstract prompts give static camera moves. Keep people small in frame or use close-ups; hands and faces in mid-shots are the weakest area of 2B-class models.
- Wan's native size is 832x480; other sizes work but cost more time. It has no image-to-video in this 1.3B variant.
- 4K from a 768x512 clip is a 5x enlargement by a super-resolution network, not native detail. It is sharp and clean (Real-ESRGAN's video network, chosen over the x4plus image network because it is about 5x faster per frame and steadier between frames) but cannot invent texture that the 2B model did not render; generate at 1024x576 first when the subject allows.
- MusicGen is instrumental only and best under 30 s; describe instruments and tempo rather than song titles or artists.